SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

Duration : 45 Minutes
QUESTIONS: 2 **Total Marks:20**

Annexure A

Descriptive Written Test

Code of Conduct:
karthikeyan/ 192419048
I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question 1

A Smart Home Automation System Using Intelligent Agents

Introduction

Artificial Intelligence (AI) enables machines to perform tasks that normally require human intelligence, such as learning, reasoning, decision-making, and problem-solving. One of the most practical applications of AI is a **Smart Home Automation System**, where intelligent agents monitor the environment and automatically control household devices.

A smart home consists of various sensors such as **motion sensors, temperature sensors, light sensors, smoke detectors, and security cameras**. These sensors continuously collect information from the environment. Based on this information, intelligent agents analyze the current situation and control devices such as lights, fans, air conditioners, alarms, smart locks, and surveillance systems through actuators.

The primary objectives of a smart home automation system are:

- Improve user comfort
- Reduce electricity consumption
- Enhance home security
- Learn user preferences
- Respond automatically to environmental changes

To achieve these objectives, different types of intelligent agents can be used.

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1. Simple Reflex Agent

Definition

A **Simple Reflex Agent** is the most basic type of intelligent agent. It makes decisions only by considering the **current percept (current sensor input)** without remembering previous events.

It follows predefined **Conditions–Action Rules**.

General Rule

IF Condition occurs
THEN Perform Action

Working in Smart Home

Suppose motion is detected in the living room after sunset.

The rule becomes

IF Motion Detected AND Time = Night
THEN Switch ON Living Room Lights

Similarly,

IF Smoke Detected
THEN Activate Fire Alarm
IF Temperature > 30°C
THEN Turn ON Air Conditioner

Advantages

- Very fast response
- Easy to implement
- Low computational cost
- Suitable for emergency situations

Disadvantages

- No memory
- Cannot learn
- Cannot predict future situations
- Performs poorly in complex environments

Example

When a person enters the kitchen at night, the lights automatically turn ON without considering previous events.

2. Model-Based Reflex Agent

Definition

A **Model-Based Agent** improves upon the simple reflex agent by maintaining an **internal model** of the environment.

It remembers previous states and combines past information with current sensor inputs before making decisions.

Working

Suppose the motion sensor temporarily stops working.

Instead of immediately switching OFF the lights, the agent remembers that the room was occupied a few seconds earlier and keeps the lights ON.

The agent continuously updates information such as

- Which rooms are occupied
- Current temperature
- Last detected motion
- Door status
- Window status

Advantages

- Maintains memory
- Handles incomplete information
- More reliable
- Better decision making

Disadvantages

- Requires additional memory
- More computationally expensive

Example

The bedroom temperature is currently 24°C.The user usually sleeps at 10 PM.

The agent remembers this pattern and automatically adjusts the air conditioner before bedtime.

3. Goal-Based Agent

Definition

A **Goal-Based Agent** selects actions that help achieve a predefined goal.

Instead of simply reacting to sensor inputs, it evaluates different possible actions and chooses the one that best satisfies the goal.

Goals in Smart Home

Examples include

- Maintain room temperature at 24°C
- Minimize electricity usage
- Ensure maximum security
- Provide user comfort

Working

Suppose

Current Temperature = 32°C

Goal = Maintain 24°C

Possible actions

- Turn ON Fan
- Turn ON Air Conditioner
- Open Smart Window

The agent compares these alternatives and selects the most appropriate action.

Advantages

- Intelligent decision making
- Planning capability
- Flexible behavior
- Suitable for dynamic environments

Disadvantages

- Higher computational cost
- Slower than reflex agents

Example

If nobody is at home, the goal changes from comfort to energy saving.

The agent automatically turns OFF unnecessary appliances.

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4. Utility-Based Agent

Definition

A **Utility-Based Agent** not only achieves goals but also selects the action that provides the **highest overall benefit (utility)**.

It considers multiple factors simultaneously.

Examples include

- Comfort
- Electricity cost
- Security
- User satisfaction
- Device health

Working

Suppose the room temperature is 29°C.

Possible actions

Action	Comfort	Energy Saving	Utility
Fan	Medium	High	80
Air Conditioner	Very High	Low	70
Open Window	Low	Very High	60

The utility-based agent selects

Fan

because it provides the highest utility score.

Advantages

- Optimizes multiple objectives
- Produces better decisions
- Learns user preferences
- Maximizes overall performance

Disadvantages

- Complex implementation
- Requires utility function design
- Higher processing time

Comparison of Intelligent Agents

Agent Type	Memory	Planning	Learning	Best Use
Simple Reflex	No	No	No	Basic automation
Model-Based	Yes	Limited	No	Partially observable environments
Goal-Based	Yes	Yes	Limited	Goal achievement
Utility-Based	Yes	Yes	Yes	Complex optimization problems

Hybrid Intelligent Agent

A modern smart home generally combines all four agent types.

Reflex Agent

Provides immediate responses during emergencies.

Example:

Smoke detected → Fire alarm activated immediately.

Model-Based Agent

Maintains information about occupancy and environmental conditions.

Goal-Based Agent

Plans actions to achieve user-defined goals.

Utility-Based Agent

Optimizes comfort, electricity usage, and security simultaneously.

This combination is called a **Hybrid Intelligent Agent**.

Justification

Among all intelligent agents, the **Hybrid Intelligent Agent** is the most suitable for smart home automation.

Reasons include:

- Fast emergency response
- Ability to remember previous states
- Intelligent planning
- Learning user preferences
- Energy-efficient operation
- Improved home security
- Better user comfort
- Reduced electricity consumption

Therefore, hybrid agents provide the most reliable, intelligent, and efficient solution for modern smart home automation systems.

Conclusion

Smart home automation demonstrates the practical application of Artificial Intelligence in everyday life. Different intelligent agents perform different roles, ranging from simple reflex actions to complex utility optimization. Although each agent has its own advantages and limitations, a **Hybrid Intelligent Agent** combines the strengths of all approaches to provide maximum comfort, energy efficiency, and security. Hence, it is the most effective architecture for intelligent smart home systems.